

CYBER ELIAS ACADEMY

COURSE SUMMARY

Computer Repairs

What You Will Learn in Every Session

12 SESSIONS

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Computer Repairs — Course Summary

A quick reference of what is covered and what you will learn in each session.

QUICK NAVIGATION

Session 01 — Computer Components

Session 02 — Safety and Tools

Session 03 — Disassembly Reassembly

Session 04 — Diagnostic Workflow

Session 05 — Hardware Diagnostics

Session 06 — Software Diagnostics

Session 07 — Component Replacement

Session 08 — Cleaning Maintenance

Session 09 — OS Installation

Session 10 — Customer Service

Session 11 — Service Reports

Session 12 — Final Project Capstone

SESSION 01

Computer Components

Week 1 â€" Module: Components and Safety

WHAT YOU WILL LEARN

- ✓ **Identify and name** — all major internal components of a desktop computer â€" motherboard, CPU and its heatsink/fan, RAM, HDD, SSD, PSU, GPU, expansion cards, cooling fans, and the main ports and connectors.
- ✓ **Explain the function** — of each major component in their own words, using a familiar comparison.
- ✓ **Describe how data flows** — through a computer from a user's input to the screen, naming the components involved at each step.
- ✓ **Distinguish clearly between RAM and storage** — , and between HDD and SSD, including speed, durability and cost differences.
- ✓ **Explain the relationship** — between components â€" how they connect to the motherboard and why the system needs all of them to work.
- ✓ **Compare a desktop and a laptop** — , stating at least three differences in how components are arranged and serviced.
- ✓ **Handle components safely** — â€" holding boards by their edges, never touching gold contacts, and using anti-static protection.
- ✓ **Identify components physically** — on a real open machine or on labelled photographs and diagrams.

SESSION 02

Safety and Tools

Week 1 â€" Module: Components and Safety

WHAT YOU WILL LEARN

- ✓ **Explain ESD (electrostatic discharge)** — in plain words, describe how it damages components, and explain why damage can be invisible.
- ✓ **Demonstrate the correct ESD-safe routine** — â€" wearing and connecting an anti-static wrist strap, working on an anti-static mat, and touching bare metal before handling parts.
- ✓ **Apply electrical safety rules** — â€" unplugging from the wall, discharging residual electricity, never working inside a powered machine, and never opening the PSU.
- ✓ **Identify and correctly name the common computer repair tools** — , and state what each is used for.
- ✓ **Use a screwdriver of the correct type and size** — and explain why the wrong size damages screws.
- ✓ **Set up a safe, organised workspace** — with good lighting, screw organisation and safe handling.
- ✓ **Practise safe lifting and handling** — of computers and components.
- ✓ **State clearly when a repair is beyond their skill level** — and must be referred to a qualified technician or supervisor.

SESSION 03

Disassembly Reassembly

Week 1 â€” Module: Components and Safety

WHAT YOU WILL LEARN

- ✓ **Prepare a machine safely** — â€” unplug, discharge, set up the anti-static mat, wear the strap, and photograph the internals before starting.
- ✓ **Safely disconnect all external cables** — from the back of a desktop computer.
- ✓ **Remove the side panel** — without damaging clips or threads.
- ✓ **Disconnect internal cables** — â€” SATA data and power, front-panel, and power connectors â€” safely and without straining them.
- ✓ **Remove components in the correct order** — (expansion card, RAM, storage drives, then cooler) and set them aside safely.
- ✓ **Reinstall components in the correct order** — , applying the correct technique (for example, RAM clips snapping into place).
- ✓ **Reconnect all cables correctly** — , using photographs and labels to guide them.
- ✓ **Verify the machine powers on** — after reassembly, and know what to do if it does not.
- ✓ **Record screw locations and cable positions** — so reassembly is accurate and repeatable.

SESSION 04

Diagnostic Workflow

Week 2 â€” Module: Diagnostics

WHAT YOU WILL LEARN

- ✓ **State and follow the five-step diagnostic workflow** — â€” gather symptoms, confirm the obvious, form a hypothesis, test one thing at a time, and verify and document.
- ✓ **Gather symptoms systematically** — by asking effective diagnostic questions and observing look/listen/smell/feel.
- ✓ **Confirm the obvious first** — â€” power, cables and switches â€” before assuming a deep fault.
- ✓ **Explain POST (power-on self-test)** — and what it means when a machine fails it.
- ✓ **Interpret basic BIOS beep codes and no-display symptoms** — as diagnostic clues (recognising that codes vary by maker and must not be invented).
- ✓ **Use swap-testing** — to isolate a faulty component safely.
- ✓ **Apply a safe boot troubleshooting order** — for common "no display" and "no power" situations.
- ✓ **Document each step** — of a diagnosis clearly, including what was tested, the result, and the conclusion.
- ✓ **Explain why a structured method beats guessing** — , using the two ideas of "likely cause" and "confirmed cause."

SESSION 05

Hardware Diagnostics

Week 2 â€" Module: Diagnostics

WHAT YOU WILL LEARN

- ✓ **Explain** — , in plain language, what a hardware fault is and how it differs from a software fault.
- ✓ **Describe** — the common symptoms of a faulty RAM stick â€" failure to boot, beep codes, random restarts, blue screens â€" and run a memory test using Windows Memory Diagnostic or MemTest86.
- ✓ **Read** — a storage drive's S.M.A.R.T. health in CrystalDiskInfo and decide whether the drive is Good, Caution or Bad; state when to back up and when to replace.
- ✓ **List** — the basic checks for a power supply fault (no power, random shutdowns) and explain why the PSU must never be opened.
- ✓ **Identify** — an overheating problem by checking temperatures with HWMonitor and inspecting dust, fans and thermal paste.
- ✓ **Diagnose** — a "no signal" display fault by testing the cable, the monitor, the GPU and the RAM in a sensible order.
- ✓ **Apply** — the swap-testing and isolation method to prove which component is faulty.
- ✓ **Record** — findings on a Diagnostic Report Form using clear, plain language a customer can understand.

SESSION 06

Software Diagnostics

Week 2 â€" Module: Diagnostics

WHAT YOU WILL LEARN

- ✓ **Explain** — the difference between a software fault and a hardware fault, and give an example of each.
- ✓ **List** — the common software faults â€" slow performance, malware infection, boot failure, driver issues, full disk â€" and match each to its usual symptoms.
- ✓ **Use Task Manager** — to identify programs using too much CPU, memory or disk, and safely end a misbehaving task.
- ✓ **Use Event Viewer** — to find error events and note their codes.
- ✓ **Run `sfc /scannow`** — in an Administrator Command Prompt to check and repair system files.
- ✓ **Run Disk Cleanup** — to free storage space safely.
- ✓ **Describe** — the safe approach to malware removal, including when to use a full scan and when a second-opinion scanner is appropriate.
- ✓ **Use System Restore / recovery options** — to return a machine to a previous working state when appropriate.
- ✓ **Decide** — whether a fault is more likely software or hardware, and explain the reasoning.

SESSION 07

Component Replacement

Week 3 â€" Module: Repair and Maintenance

WHAT YOU WILL LEARN

- ✓ **Explain** — why a replacement part must be **compatible** with the machine, and check compatibility for RAM (generation, speed, size), storage (interface) and PSU (wattage and connectors).
- ✓ **State** — the difference between a **hard disk drive (HDD)** and a **solid-state drive (SSD)**, and between **SATA** and **NVMe**, and choose the correct one for a given machine.
- ✓ **Upgrade RAM** — correctly â€" match the type, seat the stick by the notch, and confirm the new capacity is recognised.
- ✓ **Replace a storage drive** — by removing the old one, fitting the new one, connecting the correct cables, and either **cloning** the old drive or **installing Windows** on the new drive.
- ✓ **Explain "clone" and "fresh install"** — , including when each is appropriate and why backing up comes first.
- ✓ **Replace a power supply safely** — using a known-good unit of suitable wattage, reconnecting every cable and double-checking before power-on.
- ✓ **Re-apply thermal paste** — correctly (clean, apply a small amount, refit the cooler) and confirm improved cooling.
- ✓ **Verify** — a machine after any replacement using the tools from Sessions 5 and 6 (POST, Windows, Task Manager, HWMonitor).
- ✓ **Follow safety and ESD procedures** — at every step, and know when a task should be handled by a qualified technician.

SESSION 08

Cleaning Maintenance

Week 3 â€" Module: Repair and Maintenance

WHAT YOU WILL LEARN

- ✓ **Explain** — how dust, heat and blocked airflow damage a computer, and how these lead to slowness, shutdowns and part failure.
- ✓ **List** — the correct cleaning tools and **state** which tools are unsafe (e.g., ordinary vacuum cleaners).
- ✓ **Perform** — a safe internal clean â€" powering off, unplugging, using ESD care, and using compressed air, brushes and isopropyl alcohol correctly.
- ✓ **Clean** — fans, heatsinks, the case interior, keyboard, ports and screens safely and appropriately for each surface.
- ✓ **Explain** — when to re-apply thermal paste and how often, and recognise the signs (rising temperatures, dust-clogged heatsink).
- ✓ **Describe** — the environmental threats to a computer â€" dust, humidity, heat and power surges â€" and how a **surge protector** and a **UPS** help.
- ✓ **Create** — a practical **preventive maintenance schedule** for a home, a small office and a cyber cafÃ©, including who does what and how it is documented.
- ✓ **Produce** — a written **maintenance recommendation** for a real machine after cleaning and a cooling check.

SESSION 09

OS Installation

Week 3 â€" Module 3: Repair and Maintenance

WHAT YOU WILL LEARN

- ✓ **Explain in plain English** — what an operating system is and why a corrupted OS sometimes has to be reinstalled rather than repaired.
- ✓ **Back up a customer's data correctly** — before any wipe, and explain why data safety comes before every other step.
- ✓ **Check whether a machine meets the minimum hardware requirements** — for a given Windows version and say what happens if it does not.
- ✓ **Create a bootable Windows USB drive** — using the official Microsoft Media Creation Tool, in the correct order.
- ✓ **Enter the BIOS/UEFI setup or the one-time boot menu** — and set or select the correct boot device.
- ✓ **Perform a clean install of Windows** — , correctly choosing "Custom: Install Windows only (advanced)" and installing onto unallocated space.
- ✓ **Explain in beginner terms** — why we partition a drive and what a clean install removes.
- ✓ **Install drivers in the correct order** — (starting with the network driver) and run Windows Update until no updates remain.
- ✓ **Activate Windows** — using only a legitimate product key or digital licence, and state the academy's rule against piracy.
- ✓ **Restore the user's data** — and complete a **post-install verification** checklist (Device Manager, display, sound, network, essential software).

SESSION 10

Customer Service

Week 4 â€" Module 4: Professional Practice

WHAT YOU WILL LEARN

- ✓ **Greet a customer professionally** — and create a calm, trustworthy first impression at a repair shop counter.
- ✓ **Capture a fault accurately** — in the customer's own words, using the Customer Intake Form, without leading or interrupting.
- ✓ **Ask good diagnostic questions** — (what, when, how, what changed) that move a vague complaint toward a real fault.
- ✓ **Explain faults and fixes in plain language** — , avoiding jargon or translating it into everyday Nigerian English.
- ✓ **Manage expectations** — about time, cost and uncertainty honestly, including saying "I don't know yet" professionally.
- ✓ **Quote a repair and obtain written approval** — before starting any billable work.
- ✓ **Handle an unhappy or angry customer** — calmly using listen â†' empathise â†' solve â†' boundary.
- ✓ **Protect customer confidentiality** — , in the shop and outside it.
- ✓ **Build trust and repeat business** — through honesty, follow-up and consistent service.

SESSION 11

Service Reports

Week 4 â€" Module 4: Professional Practice

WHAT YOU WILL LEARN

- ✓ **Explain why service reports matter** — â€" proof of work, warranty, records and professionalism.
- ✓ **List and describe every required section** — of a professional service report.
- ✓ **Record the reported fault accurately** — in the customer's own words alongside the observed facts.
- ✓ **Document the diagnosis** — â€" tests performed, results, and the confirmed root cause.
- ✓ **Describe the action taken** — â€" parts used (with details), software installed, and work performed.
- ✓ **Record the tests performed after repair** — and the verified result.
- ✓ **Write clear, specific and honest language** — , avoiding vague phrases and unnecessary jargon.
- ✓ **Recommend preventive maintenance** — and next steps for the customer.
- ✓ **State costs, warranty terms and get the customer's signature** — on the finished report.
- ✓ **Handle customer data responsibly** — while keeping and storing records.

SESSION 12

Final Project Capstone

Week 4 â€" Module 4: Professional Practice (Capstone)

WHAT YOU WILL LEARN

- ✓ **Receive a faulty machine and complete a customer intake** — , capturing the reported fault accurately.
- ✓ **Diagnose the fault** — using the systematic diagnostic workflow (observe, ask, hypothesise, test, resolve).
- ✓ **Apply correct safety procedures** — (ESD protection, power isolation, safe tool use) throughout.
- ✓ **Perform the repair** — â€" replacement, cleaning or OS task â€" safely and correctly.
- ✓ **Verify the fix** — with real tests and confirm the machine is fully functional.
- ✓ **Produce a complete professional service report** — documenting the entire job.
- ✓ **Present and defend their work** — in a short viva/demonstration, answering questions about their decisions.
- ✓ **Reflect honestly** — on their competence, their challenges and their next steps as a technician.